

Physics Study Notes (PH201)

Academic Insight - Public Sample Revision Summary

Module 1: Maxwell Equations & EM Wave Propagation

- Gauss Laws: $\text{div}(\mathbf{D}) = \rho_v$, $\text{div}(\mathbf{B}) = 0$
- Faraday & Ampere-Maxwell: $\text{curl}(\mathbf{E}) = -\frac{d\mathbf{B}}{dt}$, $\text{curl}(\mathbf{H}) = \mathbf{J} + \frac{d\mathbf{D}}{dt}$
- Free-space wave speed $c = 3e8$ m/s, Intrinsic impedance $\eta_0 = 377$ Ohms

Module 2: Quantum Mechanics & 1D Potential Well

- De Broglie wavelength: $\lambda = h / p$
- Quantized energy in 1D box of width L : $E_n = (n^2 h^2) / (8 m L^2)$
- Note: Sample educational revision sheet provided by Academic Insight.